COURSE · VIDEO 1

Google ADK: Agents & Multi-Agent Systems

What agents are, why frameworks exist, and where Google’s Agent Development Kit fits

What is an agent?

A model that takes actions in a loop toward a goal: it decides what to do, observes the result, and decides again until the goal is met.

Tools

Acts on the world: searches the web, calls APIs, reads and writes files, runs code.

Autonomy

Chooses its own sequence of steps to complete the task.

Feedback loop

Observes each outcome and adjusts its plan before the next step.

LLM = brain · Tools = hands · Memory = recall

Multi-agent systems

Several specialist agents, each with its own LLM, tools, memory and loop, split one complex task.

Benefits

- **Focus:** narrow role, fewer tools
- **Context:** each agent has its own window; only summaries return
- **Speed:** subtasks run in parallel
- **Objectivity:** a separate reviewer has no bias toward the work

Trade-offs

- **Complexity:** harder architecture
- **Cost:** more model calls and tokens
- **Coordination:** repeated work, missed details
- **Debugging:** many agents touching shared state

Five common patterns

Pattern	How it works	Example
Orchestrator–worker	Lead agent splits the task, workers run, lead merges	Research blog from four topics
Sequential pipeline	Agents run one after another	Research → write → edit
Parallel	Subagents run at once; results merged or voted	Plagiarism, marks and profanity checks
Debate and critic	Proposer and critic iterate until the critic approves	Coder and code reviewer
Peer collaboration	Agents talk freely with no single boss	Simulated agent village

Why use a framework?

Control flow

Retries, timeouts, fallbacks, MCP tools and memory become heavy plumbing code.

Orchestration

Handoffs, shared state, parallel work and merged results are hard to hand-write.

Model-agnostic

Message, tool-calling and streaming formats differ by provider; a framework hides them.

Alternatives: LangGraph (most mature, low-level control), CrewAI, Agno, PydanticAI, OpenAI Agents SDK, Microsoft Agent Framework, Strands, Claude Agent SDK.

Hand-code to learn; use a framework for complex, production systems

Google ADK

Open-source, code-first Agent Development Kit. Built inside Google, released April 2025 with A2A and Agent Engine; ADK 2 (2026) adds graph workflows.

Strengths

- Built for multi-agent systems
- Works with any model; best with Gemini
- Python, TypeScript, Java, Go, Kotlin
- Native A2A, web UI, built-in evaluation
- Structured sessions, memory and artifacts

Trade-offs

- Opinionated project structure and lifecycle
- Pull toward Google models and Cloud

Course roadmap

1. Setup and first agent
2. Tools
3. Structured output
4. ADK internals
5. MCP
6. Sessions, state, memory
7. RAG agent

1. Sequential, parallel, loop workflows
2. First multi-agent system
3. Graph workflows (ADK 2)
4. A2A
5. Human in the loop
6. Evaluation
7. Deployment

Rough plan: order and count may change